Supplementary File S7

Assessment Rubrics for All Four Lesson Plans (Instructor Use)

Indian Cuisine Classification Framework — A Culturally Contextualized Image Dataset for Statistical Learning Education

> *these rubrics are drafted directly from the "Assessment" line items in Tables 3–6 of the manuscript (Sections 6.2–6.5). Section 6.7 reports that Lesson 1 and Lesson 3 assessments were scored against rubrics during the pilot (87.5%, 83.3%, and 75% pass rates cited). If the rubric actually used during grading differs from what's drafted here — different criteria, a points scale rather than four performance levels, etc. — replace the relevant table below with the original so that S7 reflects what was actually used to produce those reported figures, rather than a reconstruction.*

Lesson Plan 1: Data Splitting and Class Imbalance

Assessment task: Short written response describing one real-world classification problem where a class-imbalanced dataset could cause harm if evaluated using accuracy alone, with a proposed and justified alternative metric.

Criterion	Exceeds Expectations	Meets Expectations	Approaching Expectations	Does Not Meet Expectations
Scenario identification	Identifies a realistic, specific real-world classification scenario where class imbalance and accuracy-only evaluation would cause tangible harm (e.g., a named domain with clear stakes).	Identifies a plausible real-world scenario where accuracy alone would be misleading, with a clear link to class imbalance.	Identifies a scenario but the connection to class imbalance or to potential harm is vague or only partially explained.	Scenario is missing, generic (not tied to classification), or does not involve class imbalance.
Alternative metric proposal	Proposes an appropriate alternative metric (e.g., per-class recall, weighted F1, macro F1) and justifies the choice with specific reasoning tied to the scenario's stakes.	Proposes a reasonable alternative metric with basic justification.	Proposes a metric but justification is weak, generic, or only loosely connected to the scenario.	No alternative metric proposed, or the proposed metric does not address the imbalance problem.
Conceptual understanding	Response demonstrates clear understanding of why overall accuracy obscures per-class performance under imbalance, using correct terminology.	Response shows adequate understanding of the accuracy-imbalance problem.	Response shows partial or somewhat confused understanding of the underlying concept.	Response shows a fundamental misunderstanding of accuracy vs. per-class metrics.

Lesson Plan 2: The Bias-Variance Trade-off in Practice

Assessment task: One-page written analysis describing the bias-variance trade-off observed in the student's own three-model experiment, a recommended deployment capacity, and one proposed additional regularization strategy.

Criterion	**Exceeds Expectations**	**Meets Expectations**	**Approaching Expectations**	**Does Not Meet Expectations**
Description of observed trade-off	Accurately describes the bias-variance trade-off observed across the three model capacities, citing specific train/val accuracy figures from their own experiment.	Describes the trade-off with general reference to their results, mostly accurate.	Describes the trade-off in general terms without clearly connecting it to their own experimental results.	Does not describe the trade-off, or description is factually incorrect.
Model capacity recommendation	Recommends a specific model capacity for deployment with well-reasoned justification weighing accuracy, generalization gap, and practical constraints.	Recommends a model capacity with reasonable justification.	Recommends a model capacity but justification is thin or one-sided (e.g., considers only accuracy).	No clear recommendation, or recommendation contradicts their own reported results.
Regularization proposal	Proposes an additional regularization strategy not already used (e.g., dropout, weight decay, early stopping) and correctly explains its expected effect on the bias-variance trade-off.	Proposes a valid regularization strategy with a basically correct explanation.	Proposes a strategy but the explanation of its effect is vague or partially incorrect.	No strategy proposed, or the proposed strategy is not a regularization technique.

Lesson Plan 3: Confusion Matrix and Multiclass Evaluation

Assessment task: For the three dish categories with the lowest recall identified by the student, 2–3 sentences per class explaining (a) the likely visual reason for low recall and (b) one data-collection or modeling change that might improve it.

Criterion	Exceeds Expectations	Meets Expectations	Approaching Expectations	Does Not Meet Expectations
Visual feature reasoning (per class, x3)	For all three selected low-recall classes, offers a specific, plausible visual explanation (shared color, shape, texture, or plating) grounded in the confusion matrix / example images.	For most of the three classes, offers a plausible visual explanation with reasonable specificity.	Explanations are present but generic (e.g., 'they look similar') without identifying specific shared features.	No visual explanation offered, or explanation is unrelated to the actual confusion pattern in the data.
Corrective measure proposal (per class, x3)	For all three classes, proposes a specific, actionable data-collection or modeling change (e.g., targeted data collection, class-specific augmentation, hierarchical classification) with justification.	Proposes reasonable corrective measures for most classes.	Proposes measures but they are generic (e.g., 'collect more data') without being tailored to the specific confusion pattern.	No corrective measure proposed, or proposal is not actionable.
Use of evaluation metrics	Correctly distinguishes precision, recall, and F1 in the write-up, and uses the correct metric to justify the choice of 'low recall' classes.	Uses metrics correctly with minor imprecision in terminology.	Conflates precision and recall, or uses metric terminology loosely.	Does not engage with the specific metrics; relies only on overall accuracy.

Lesson Plan 4: Data Augmentation as Regularization

Assessment task: Design an augmentation pipeline for a different image classification problem of the student's choice, justifying each included transformation and identifying one transformation to explicitly avoid.

Criterion	Exceeds Expectations	Meets Expectations	Approaching Expectations	Does Not Meet Expectations
Augmentation pipeline design	Designs a coherent augmentation pipeline for the chosen new domain (e.g., medical or satellite imagery), selecting transformations appropriate to that domain's visual properties.	Designs a reasonable pipeline with mostly appropriate transformations.	Pipeline is present but includes transformations poorly matched to the chosen domain.	No coherent pipeline proposed, or transformations are copied from the food dataset without adaptation.
Justification of included transformations	Each included transformation is justified with reference to what real-world variation it simulates and why the class label is preserved under it.	Most transformations are justified with reasonable reference to label preservation.	Justifications are present but generic or do not address label preservation.	No justification provided for included transformations.
Identification of an inappropriate transformation	Identifies a specific transformation that would be inappropriate for the chosen domain and explains precisely why it would change the effective label or introduce unrealistic artifacts (e.g., horizontal flip on a chest X-ray, or rotation on a satellite image with a fixed north reference).	Identifies an inappropriate transformation with a generally correct explanation.	Identifies a transformation to avoid but the reasoning is vague or partially incorrect.	No transformation identified as inappropriate, or the example given does not actually pose a problem.

Scoring Guidance

Each lesson's assessment has 3 criteria. Suggested translation to a numeric scale for gradebook purposes, if needed:

Exceeds Expectations = 4 pts | Meets Expectations = 3 pts | Approaching Expectations = 2 pts | Does Not Meet Expectations = 1 pt

Total possible per lesson assessment: 12 points (3 criteria × 4 pts). Each lesson's assessment is graded on its own, and the Section 6.7 pilot reports pass rates per lesson, so apply the threshold per lesson: a score of ≥ 9/12 (i.e. "Meets Expectations" on average across the three criteria) is a reasonable pass mark for the pass/fail framing used in Section 6.7 (e.g., "correctly identified...with valid reasoning"). Aggregated across all four lesson assessments, that threshold is ≥ 36/48. Adjust thresholds to match institutional grading conventions.